

Here is the same device used in Lemma VII. We produce AD to the remote point d and AE to e , keeping $Ad : Ae = AD : AE$, and draw the ordinates db and ec parallel to the original BD and CE . A similar curve Abc is drawn out at this fixed distance, and the tangent Ag touches both curves at A , cutting the four ordinates in F, G, f, g . The genius of the trick is scale invariance: while the original figure near A vanishes, this enlarged copy at d, e stays a fixed finite size, so we may take limits there in plain view. Because every length is proportional, the areas $\triangle ABD$, $\triangle ACE$ stay in constant ratio to the blown-up areas Abd , Ace .